

Radar Research for ProofRoad

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Why radar is the primary proof sensor

ProofRoad assigns radar the unique authority to prove the road clear because it is expected to be less affected by fog, rain and dust than the optical modalities selected for the prototype. It also acknowledges radar weaknesses: object naming, pit-wall clutter, reflections and coarse angle (pp. 3–4).

The proposal specifies 77 GHz for a mine implementation and a 60 GHz evaluation board or stand-in for the demo (p. 4).

2. Radar processing pipeline

1) Acquire range/velocity detections. 2) Filter invalid/noisy returns. 3) Track detections across frames. 4) Associate tracks with the mapped route/corridor. 5) Estimate nearest relevant hazard. 6) Feed the safety MCU's proven-distance calculation. 7) Log raw decision inputs and resulting state.

Association is rule-based rather than learned in the proposal. AI on the assist computer can name what sensors found, but it does not get authority to clear a hazard (p. 4).

3. Radar validation plan

Test stationary and moving metal targets at known distances.

Repeat under clear, artificial mist, rain-like scattering and dust conditions, while explicitly labelling artificial conditions as non-equivalent to mine fog.

Measure detection probability, false alarms, range error, velocity error, latency, clutter behavior and failure states.

Test radar interference and degraded modes. ProofRoad lists radar interference as a field-test item and requires degraded-state behavior when radar is blocked (p. 10).

4. External research reference

The RADIATE dataset paper, presented at ICRA 2021, is directly relevant to radar perception in bad weather and can be used as a research reference for adverse-weather radar perception. It is not a mine-specific dataset and therefore should not be treated as direct validation for ProofRoad.

Reference: Sheeny et al., RADIATE: A Radar Dataset for Automotive Perception in Bad Weather, ICRA 2021, DOI 10.1109/ICRA48506.2021.9562089.

5. Radar research questions

What minimum radar range is required to maintain useful safe-speed advice at the mine's actual maximum speed?

How should radar confidence/health be represented without turning it into an unsupported percentage safety score?

How should wall/rock clutter be rejected without filtering out a real stopped vehicle?

What mounting height, tilt and field of view are appropriate for the actual dumper geometry?